The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity

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Abstract

On a finite screen with fixed capacity, maximizing generalized entropy over sector probabilities gives $S_{\rm gen}^{\rm max} = \log M$, and uniform transfer of a fraction f of the horizon capacity to an observer changes both this maximum and the logarithmic sector observable $\mathcal{A}$ by $\log(1-f) < 0$. The unperturbed horizon is therefore a one-sided boundary maximum along the transfer direction. The exact gradient and Hessian of $\mathcal{A}$ show that this statement cannot be replaced by an interior-maximum claim for that observable: at fixed horizon capacity its tangent Hessian is positive, while its homogeneous curvature is negative, and the joint counting entropy is a nonstrict maximum. Under a horizon-capacity dictionary this fixed-budget law supplies a conditional sign mechanism for the time advance and increasing regularized out-of-time-ordered correlator of negative de Sitter shocks, a sign that conflicts with the positivity and cyclicity required by a proposed static-patch trace. For an icosahedral screen, a normalized finite shock spectrum follows conditionally from exact gauge transport of the rotation triplet and a nearest-neighbour Laplacian identification, with the pure-de-Sitter normalization and the line-graph spectrum identity exact. The calculation does not derive the coefficient or gravitational attachment and supplies no static-patch trace.

Keywords: de Sitter horizon, shock wave, static patch, finite screen, generalized entropy, icosahedral graph

1 The sign problem

Chen, Stanford, Tang, and Yang study a static-patch proposal in which a Euclidean gravitational path integral with an observer worldline computes a Hilbert-space trace [1]. Their shock-wave calculation includes observer recoil and gravitational backreaction. The resulting out-of-time-ordered correlator conflicts with cyclicity and positivity: its leading regularized contribution increases. Geometrically, the negative shock advances the signal in the horizon null coordinate.

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The finite-screen calculation below isolates a capacity mechanism for this sign. Three levels are kept separate throughout:

1. exact entropy and graph identities for a finite sector model;
2. an analytic relaxation of integer sector dimensions for stating gradients and curvatures;
3. the physical dictionary identifying screen capacity transfer with a gravitational horizon shock.

The first two levels are mathematical statements. The third is a conditional attachment. In particular, a finite capacity premise does not evade the trace obstruction in Ref. [1]. Finite-dimensionality is part of the premise tested there.

2 The smooth de Sitter normalization

Write the horizon shock equation in spacetime dimension d as

$$\left(-\nabla^2_{S^{d-2}} - \mu^2\right) X^+ = 8\pi G\, T^-, \qquad \mu^2 = \frac{d-2}{2}|f'(r_c)|r_c = (d-2)\kappa r_c. \tag{1}$$

For pure de Sitter space of radius L,

$$f(r) = 1 - \frac{r^2}{L^2}, \qquad r_c = L, \qquad \kappa = \frac{1}{L}.$$

Lemma 2.1 (Radius-independent shock mass)**.** *For pure de Sitter space,*

$$\mu^2 = d - 2 = \lambda_{\ell=1}\left(S^{d-2}\right).$$

Consequently, μ^2 is independent of L and of the cosmological constant.

Proof. The product κr_c equals one. Equation (1) therefore gives $\mu^2 = d-2$. Scalar spherical harmonics on S^{d-2} have

$$\lambda_\ell = \ell(\ell + d - 3),$$

so $\lambda_1 = d - 2$. □

The smooth $\ell = 1$ shock is generated by a de Sitter isometry and is pure gauge, with the corresponding source component removed by the gravitational constraint [1]. This explains the zero of the operator in Eq. (1). Transferring that normalization to a discrete screen requires a separate premise, stated in Section 4.

3 Finite entropy and the capacity-transfer direction

3.1 The exact entropy maximum

Let a finite screen have positive integer sector dimensions $d_1, \ldots, d_n$, and let p_i be a probability distribution over the sectors. Set

$$M = \sum_{i=1}^n d_i, \qquad S_{\mathrm{gen}}(p, d) = -\sum_{i=1}^n p_i \log p_i + \sum_{i=1}^n p_i \log d_i.$$

This is the counting entropy of sectors with one-dimensional bulk blocks, or with d_i counting the whole sector. For blocks $\rho = \bigoplus_i p_i(\rho_i \otimes I_{d_i}/d_i)$ with bulk dimensions b_i, the full entropy is $\log M - D_{\mathrm{KL}}(p\|w/M) - \sum_i p_i D_{\mathrm{KL}}(\rho_i\|I_{b_i}/b_i)$ with $w_i = b_i d_i$ and $M = \sum_i w_i$, so saturation also requires maximally mixed bulk blocks [3].

Proposition 3.1 (Finite sector entropy maximum)**.** *For fixed d_i,*

$$p_i^\star = \frac{d_i}{M}, \qquad \max_p S_{\mathrm{gen}}(p,d) = \log M.$$

Proof. With $q_i = d_i/M$,

$$S_{\mathrm{gen}}(p,d) = \log M - \sum_i p_i \log \frac{p_i}{q_i} = \log M - D_{\mathrm{KL}}(p\|q).$$

Nonnegativity of relative entropy proves the result. □

Thus a logarithmic capacity $N = \log M_0$ agrees exactly with the extremal generalized entropy when the closed screen carries total dimension M_0. The same identity compares every admissible integer partition with total count M: $S_{\mathrm{gen}}(p,d) \le \log M$, with equality exactly at $p = d/M$. In the positive-real relaxation, every twice differentiable joint path at fixed M through $p = d/M$, with $u = p'(0)$ and $v = d'(0)$, has $S'_{\mathrm{gen}}(0) = 0$ and $S''_{\mathrm{gen}}(0) = -\sum_i (u_i - v_i/M)^2/p_i \le 0$, so saturation is a nonstrict joint maximum at fixed M [3]. Neither statement constructs an operation that changes integer sector dimensions.

3.2 Exact gradient and Hessian

At the maximizing distribution, the expectation of the logarithmic sector operator is

$$\mathcal{A}(d) := \sum_{i=1}^{n} \frac{d_i}{M} \log d_i. \tag{2}$$

The dimensions are integers in the finite model. Extend them to positive real variables solely for the following differential calculation.

Proposition 3.2 (Analytic curvature of the sector observable)**.** *The exact gradient of Eq. (2) is*

$$\frac{\partial \mathcal{A}}{\partial d_i} = \frac{\log d_i + 1 - \mathcal{A}}{M}. \tag{3}$$

Writing $g_i = \partial_i \mathcal{A}$, its Hessian is

$$\frac{\partial^2 \mathcal{A}}{\partial d_i \partial d_j} = \frac{\delta_{ij}}{M d_i} - \frac{g_i + g_j}{M}. \tag{4}$$

At the symmetric point $d_i = d$,

$$\mathrm{Hess}\,\mathcal{A} = \frac{1}{nd^2}\left(I - \frac{2}{n}J\right), \tag{5}$$

where $J = \mathbf{1}\mathbf{1}^T$. Its eigenvalue on $\mathbf{1}$ is $-1/(nd^2)$, and every vector orthogonal to $\mathbf{1}$ has eigenvalue $+1/(nd^2)$.

Proof. Differentiate $\mathcal{A} = M^{-1}\sum_i d_i \log d_i$ to obtain Eq. (3). A second derivative gives Eq. (4). At $d_i = d$, one has $M = nd$, $\mathcal{A} = \log d$, and $g_i = 1/(nd)$, which yields Eq. (5). □

Two consequences prevent a false extremum statement. First, $\nabla\mathcal{A} = \mathbf{1}/(nd)$ at the symmetric point, so the point is not stationary in the full positive orthant. Second, a fixed-horizon-capacity variation obeys $\mathbf{1}^T\delta d = 0$, and hence

$$\delta d^T(\mathrm{Hess}\,\mathcal{A})\delta d = \frac{\|\delta d\|^2}{nd^2} > 0$$

for every nonzero tangent variation. The symmetric point is locally minimizing along the fixed-M redistribution directions. The negative eigenvalue belongs to the homogeneous direction, which changes M.

3.3 The one-sided budget maximum

The de Sitter sign concerns transfer between the horizon and the observer, rather than redistribution among horizon sectors. Let the closed total capacity be M_0. Assign $c_{\mathrm{obs}} \geq 0$ units in the finite capacity ledger to the observer, leaving

$$M_h = M_0 - c_{\mathrm{obs}}$$

on the horizon. Suppose the transfer depletes every horizon sector by the same factor $1 - f$, where $f = c_{\mathrm{obs}}/M_0$. For integer dimensions, restrict to values for which the depleted dimensions remain integers. Then

$$S^{\max}_{\mathrm{gen},h}(f) - S^{\max}_{\mathrm{gen},h}(0) = \log(1-f), \tag{6}$$

and

$$\mathcal{A}((1-f)d) - \mathcal{A}(d) = \log(1-f). \tag{7}$$

Both relations are exact.

Theorem 3.3 (Finite transfer law and analytic one-sided maximum)**.** *Every admissible positive integer transfer has a smaller horizon entropy maximum and uniformly depleted sector observable $\mathcal{A}$ than the zero-transfer allocation. On the positive-real interpolation $0 \leq c_{\mathrm{obs}} < M_0$, the change is strictly decreasing and concave:*

$$\frac{d}{dc_{\mathrm{obs}}}\log(M_0 - c_{\mathrm{obs}}) = -\frac{1}{M_0 - c_{\mathrm{obs}}} < 0, \qquad \frac{d^2}{dc_{\mathrm{obs}}^2}\log(M_0 - c_{\mathrm{obs}}) = -\frac{1}{(M_0 - c_{\mathrm{obs}})^2} < 0.$$

The allocation $c_{\mathrm{obs}} = 0$ is a one-sided boundary maximum on both the admissible finite transfers and the analytic transfer path.

For transfers of 2%, 5%, and 10%,

$$\log(0.98) = -0.020203\ldots, \qquad \log(0.95) = -0.051293\ldots, \qquad \log(0.90) = -0.105361\ldots.$$

The exact finite order and its analytic interpolation are the budget statements relevant to the shock sign. Proposition 3.2 supplies compatible negative homogeneous curvature, while also showing why the transfer result cannot be described as an interior maximum of $\mathcal{A}$ at fixed horizon M.

3.4 Conditional gravitational interpretation

In the horizon coordinates used in Ref. [1], the constant term of the shock operator carries the sign of $B'(0)/B(0) = 2r'(0)/r_c$. For a cosmological horizon, the transverse area decreases from the horizon toward the observer, and the shock has the time-advance sign. This radial profile belongs to the gravitational dictionary. The native screen energy excludes a boundedly invertible L^2 energy congruence with the boundary action of a smooth compact three-dimensional Dirichlet bulk, so a bulk reading of the screen needs a radial observation algebra that the native energy does not supply [6]. Under the two physical identifications

$$\text{closed screen capacity} = \text{fixed horizon–observer budget}, \qquad \mathcal{A} = \frac{A_{\text{horizon}}}{4G},$$

Theorem 3.3 supplies the corresponding conditional sign mechanism. Identifying the capacity coordinate c_{obs} with the observer mass used in the gravitational shock calculation is an additional physical dictionary. The finite entropy theorem and transfer law establish none of these identifications. On the executable source, fixed total count and $\log d$ have no established physical volume or area map. Its full return reference has entropy deficit at most $2025/(64V)$ for V ports, and on a refining base-face patch its mutual information per crossing seam tends to zero, so no nonzero universal coefficient multiplies boundary-seam counts of regions on the two-dimensional source [3]. The native regional algebra of one seam does have the sector form of this section, with exterior multiplicities $45, 10, 1$ and an exact bulk/edge entropy split, and a fixed-occupation filter there has nonzero entropy slope at constant bare cut count, so those identities supply neither responsive area nor fixed-proper-volume stationarity [5]. The identification above reads the whole screen count, which those controls leave open.

4 Conditional icosahedral spectrum

Consider the graph Laplacian of the twelve vertices of the icosahedron. Its eigenspaces decompose under the alternating group A_5 on five letters as

$$\mathbb{R}^{12} \simeq \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5} \oplus \mathbf{3}',$$

with raw Laplacian eigenvalues

$$0, \qquad 5 - \sqrt{5}, \qquad 6, \qquad 5 + \sqrt{5},$$

respectively. This graph is the seam graph of the twelve-port carrier of the executable observer. Under its uniform independent seam schedule both native repairs have quantum spectral gap $(5 - \sqrt{5})/60$ per attempt; that gap describes mixing on a fixed carrier, not a shock spectrum [8].

Assumption 4.1 (Discrete gauge transport DS-GAUGE)**.** *The* $\mathbf{3}$ *generated by the screen isometries consists of exact gauge zero modes of the discrete shock operator, in direct analogy with the smooth* $\ell = 1$ *isometry modes.*

Assumption 4.2 (Nearest-neighbour kinetic identification DS-LAPLACIAN)**.** *The discrete shock kinetic operator is the nearest-neighbour combinatorial Laplacian on the port graph, or its edge-sector line graph, with one positive overall scale.*

These two assumptions are independent. Assumption 4.1 fixes the scale by placing the **3** at the smooth value 2. Assumption 4.2 selects the finite operator whose remaining eigenvalues are being compared.

Proposition 4.3 (Conditional normalized shock spectrum)**.** *Under Assumptions 4.1 and 4.2, the normalized port Laplacian and shock spectra are*

A_5 *block*	*multiplicity*	λ_{norm}	λ_{shock}
1	1	0	-2
3	3	2	0
5	5	$3+\dfrac{3}{\sqrt{5}}$	$1+\dfrac{3}{\sqrt{5}}$
3′	3	$3+\sqrt{5}$	$1+\sqrt{5}$

The invariant mode agrees exactly with the smooth value -2*. The* **5** *shock eigenvalue is* $1+3/\sqrt{5} = 2.341640\ldots$*, and the additional* **3′** *triplet lies at* $1+\sqrt{5} = 2\varphi = 3.236067\ldots$*.*

Proof. Multiply the graph Laplacian by $2/(5-\sqrt{5})$, then subtract $\mu^2 = 2$. Direct simplification gives the displayed values. □

The value -2 carries no enhancement relative to the continuum invariant mode. Any different comparison obtained by leaving the graph scale unfixed is a normalization artifact.

4.1 Ports and edge sectors

The logarithmic sector operator naturally suggests an edge carrier. The low-lying spectrum is unchanged by this choice.

Lemma 4.4 (Regular line-graph spectrum)**.** *Let G be a k-regular graph with $k \geq 2$, n vertices, and m edges. The line graph $L(G)$ has the Laplacian spectrum of G, together with the eigenvalue $2k$ repeated $m-n$ times.*

Proof. Let B be the unoriented vertex-edge incidence matrix. Then

$$BB^T = A_G + kI, \qquad B^TB = A_{L(G)} + 2I.$$

The nonzero spectra of BB^T and B^TB agree, and the spectrum of the $m \times m$ matrix B^TB contains $m-n$ additional zeros relative to the spectrum of the $n \times n$ matrix BB^T, counted with multiplicity. Since $L(G)$ is $(2k-2)$-regular, converting adjacency eigenvalues to Laplacian eigenvalues gives the claim. □

For the icosahedron, $k = 5$, $n = 12$, and $m = 30$. The edge-sector spectrum is therefore

$$\{0^1, (5-\sqrt{5})^3, 6^5, (5+\sqrt{5})^3, 10^{18}\}.$$

It contains the full port spectrum plus a high block at 10. In particular, the normalization-free ratio between the **5** and rotation-triplet Laplacian eigenvalues is

$$\frac{\lambda_{\mathbf{5}}}{\lambda_{\mathbf{3}}} = \frac{6}{5-\sqrt{5}} = \frac{3(5+\sqrt{5})}{10} = 2.170820\ldots$$

on both carriers. The dodecahedral face graph gives $\varphi^2 = 2.618033\ldots$, while the smooth $\lambda_{\ell=2}/\lambda_{\ell=1}$ ratio is 3.

Lemma 4.4 closes the port-versus-edge ambiguity once the nearest-neighbour Laplacian premise is accepted. It does not establish that premise.

5 The repair generator has a different domain

Let X_r be the configuration space at fixed sector structure, with measure π_r, and let P_v be the single-site heat-bath conditional expectation. The declared heat-bath generator

$$L^{\mathrm{rep}} = \sum_v (I - P_v)$$

acts on

$$K_r = L^2(X_r, \pi_r).$$

Since each P_v is an orthogonal projection,

$$\langle f, L^{\mathrm{rep}} f\rangle = \sum_v \|(I - P_v)f\|^2.$$

Consequently,

$$\ker L^{\mathrm{rep}} = \bigcap_v \mathrm{Ran}\, P_v.$$

Reducing this intersection to the constants requires a separate irreducibility or intersection proof and is unnecessary for the scope statement here.

The executable native action is the seam sum $H_{\mathrm{native}} = \sum_e (I - \Phi_e)$, a positive local Liouville operator built from the record-discarded native repair Φ_e of each seam [7]. It acts on the operator algebra of a fixed carrier and conserves total occupation, whose fixed-number sectors its connected exchanges generate. It therefore changes no sector dimension either, and the domain statement below applies to it.

A horizon shock changes the sector dimensions d_α in

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha.$$

Such a change deforms the configuration space and lies outside $L^2(X_r, \pi_r)$ at fixed sector structure. Neither generator therefore acts on the sector-dimension shock. Their spectral gaps neither exclude nor supply the time-advance mode. The capacity-transfer law in Theorem 3.3 is a response input on the enlarged sector-dimension space; no dynamical generator on that space is supplied.

A concrete stationary repair control. The balanced conservative twelve-port family illustrates the domain distinction with its actual local repair law. On a connected weighted port graph, let $z = n - (r/p)\mathbf{1}$ and $\kappa = r(p-r)/[p(p-1)]$. Fair endpoint exchange at each Poisson attempt has density drift $Gz = -Lz/2$, uniform stationary occupancy, and $\mathrm{Cov}(z(t), z(0)) = \kappa e^{-|t|L/2}\Pi$. Here t is the Poisson schedule parameter, one expected attempt per seam per unit; it is a model coordinate, not a physical, modular or observer clock time. At fixed graph, its integrated fluctuation record

has covariance $4\kappa L^{+}$. This derives a graph Green covariance from local port readback and repair histories; it neither deforms sector dimensions nor identifies the record with a horizon shock.

For a stationary reversible continuous-time chain and centered current-state observable f, spectral resolution gives

$$\langle f, e^{tG} f\rangle_{\pi} = \sum_{j} a_j^2 e^{-\gamma_j t}, \qquad \gamma_j \geq 0.$$

It is nonnegative and nonincreasing. A nondecreasing global time change preserves that property. The discrete-time version requires nonnegative eigenvalues, satisfied by a lazy kernel $(I+K)/2$; the nonlazy two-state flip instead gives $(-1)^m$. These are controls for a specified classical autocorrelation, not constraints on an out-of-time- ordered quantum correlator or an enlarged sector-dimension dynamics. Exact finite witnesses and source-pinned operator controls are archived with the repository evidence [11].

The inverse covariance also retains geometric anisotropy: $[L^{-1}, U] = -L^{-1}[L, U]L^{-1}$ on the mean-zero subspace. The native level-four seven-mode band has 14.642% splitting, whereas an independently supplied round-metric control has 0.01561%. Band averages or a $3, 5, 3$ multiplicity pattern therefore do not by themselves supply the rotational covariance assumed by a round horizon attachment. The conditional shock-spectrum identification must keep its metric and representation premises.

6 The fixed-capacity equation of state

The capacity reading of the cosmological-constant term supplies a conditional branch test through the dark-energy equation of state, and its algebraic law is machine-checked. Read the dark-energy density through the capacity relation as $\rho_{\mathrm{DE}}(a) = \kappa/N(a)$, with κ the single positive constant packaging $c^4\, 3\pi/(8\pi G\, l_P^2)$, and define the effective equation of state through the FRW continuity relation, $w(a) = -1 - \frac{1}{3}\,\mathrm{d}\ln\rho_{\mathrm{DE}}/\mathrm{d}\ln a$, which is the effective-fluid quantity projected by a w_0w_aCDM analysis within its chosen cosmological likelihood. Three exact conditional statements follow, each proved in `FixedCapacityWLaw` of the Lean corpus with the capacity value cancelling throughout.

First, on the fixed-capacity branch, constant N, the equation of state is $w(a) = -1$ identically, and the CPL representation is forced to $(w_0, w_a) = (-1, 0)$ exactly, with zero continuous freedom within that branch. This is the immediate consequence of the assumed $\rho_{\mathrm{DE}} = \kappa/N$ identification and the continuity definition of w; it does not derive fixed capacity, the density identification, or a closed dark sector. Second, the exact converse holds on the drift branch: $w(a) = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so any measured departure from -1 is a capacity-drift readback only conditional on those same identifications; the power law $N \propto a^{\varepsilon}$ gives the exact line $w = -1 + \varepsilon/3$. Third, if the record capacity is monotone nondecreasing, then $w(a) \geq -1$ at every epoch: this conditional branch forbids a phantom equation of state, and an established $w(a_0) < -1$ forces capacity loss at a_0 through the strict readback.

Empirical comparison is conditional on this capacity dictionary and on the cosmological likelihood. A fixed branch and a monotone branch are different hypotheses: failure of the fixed point cannot be rescued by switching to monotonicity. Compatibility with $(-1, 0)$ is not distinctive support because it is also the standard ΛCDM point. The registered decision rules and retrospective DESI chain comparisons are given in the prediction ledger and cosmological analysis [9, 10, 2]. They are not direct measurements of capacity. The identity $N = 3\pi/(\Lambda\ell_P^2)$ converts a cosmological posterior

into a capacity posterior only after the de Sitter entropy dictionary is assumed; it does not select N.

The continuity relation defines w from a supplied density history and supplies no work measurement: consensus about identical fixed-volume records cannot select an unobserved strain derivative, since $H_\alpha(V) = Q(V/V_0)^{-\alpha}$ gives $p/\rho = \alpha$. Three geometrically spaced energy readings identify the response within a family with $\alpha \geq 0$, which excludes $w = -1$ at positive energy, and an additive energy offset changes p/ρ while leaving pressure unchanged [4].

A second conditional diagnostic composes the capacity law with the local electroweak-closure tangent. It requires three extra physical premises: (B1) the closure capacity is the same physical N; (B2) the finite closure relation is an epochwise physical law; and (B3) physical evolution selects the committed root branch and co-variation convention while the local clock reads the same homogeneous cosmological α with the required time map. The mathematical derivative is interval-certified independently of those physical premises. An outward-rounded implicit-function calculation proves the selected declared root branch continuously differentiable for $|\Delta \ln \alpha| \leq 10^{-5}$, with

$$\frac{\mathrm{d}\ln N}{\mathrm{d}\ln \alpha} \in [-0.214173865, -0.206176031], \qquad \frac{\mathrm{d}\ln \alpha}{\mathrm{d}\ln N} \in [-4.850224325, -4.669103775].$$

The reciprocal is formed only after zero is excluded. Using the conservative slope bound and $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, all three physical premises together let the optical-clock drift limit imply $|1 + w_0| \leq 6.94 \times 10^{-10}$ [13]. The static closure solver does not select the physical cosmic branch, and its certified local domain cannot be extrapolated over the order-one changes suggested by some CPL histories. The diagnostic restricts but does not empty the monotone branch.

7 Claim boundary

Statement	Scope
$\mu^2 = d - 2 = \lambda_{\ell=1}$ in pure de Sitter	exact smooth-geometry identity
$S_{\mathrm{gen}}^{\max} = \log M$	exact finite entropy maximum over sector probabilities
Uniform-depletion drop $\log(1 - f)$	exact finite law for admissible integer dimensions
Gradient and Hessian of $\mathcal{A}(d)$	exact analytic-relaxation identity
One-sided horizon-budget maximum	exact order on admissible finite transfers and exact analytic interpolation; gravitational reading conditional
Port/edge low-spectrum equality	exact regular line-graph theorem
Icosahedral shock spectrum	conditional on exact discrete gauge transport and the scaled nearest-neighbour Laplacian identification
Shock coefficient and physical screen attachment	not derived
Static-patch trace representation	excluded from the claim; the cited trace obstruction stands

The coefficient multiplying the shock response requires a generalized-entropy normalization and a

physical horizon-area dictionary on one common domain. The finite calculation fixes neither. It also gives no Hilbert-space trace whose correlators satisfy the conditions tested in Ref. [1].

Closed tested classes and surviving alternatives

The external trace obstruction concerns the proposal tested in Ref. [1]: a Euclidean static-patch path integral with an observer worldline interpreted as a positive cyclic Hilbert-space trace for the stated correlators. Horizon-screen descriptions without that trace interpretation are outside the tested class.

The repair-domain statement concerns the declared fixed-sector L^{rep} on $L^2(X_r, \pi_r)$. A separately constructed dimension-changing generator is not excluded. The graph-spectrum statement concerns the scaled nearest-neighbour icosahedral port Laplacian and its line graph under exact gauge transport. Weighted, longer-range, refinement-limit, and other source-derived kinetic operators are possible and require their own spectra. One refinement-limit operator is constructed: along a common subsequence of refining source screens the native cell Green operators converge in operator norm to a compact injective G, and $A = G^{-1}$ defines a regular, strongly local Dirichlet form with $n/49920 \le \lambda_n(A) \le 3n/2$ and heat-trace spectral dimension two [6]. That limit selects no unique whole-sequence limit and no physical metric, so a shock reading of A needs its own gauge-transport and normalization premises. On the executed federated screens of 240, 960 and 3,840 ports, the numerically computed ratio of the second to the first low band of the native port Laplacian is 2.68, 2.92 and 2.97, against $6/(5-\sqrt{5}) \approx 2.17$ on one carrier and 3 on the round sphere [12]. The lowest band is a near triplet whose relative spread grows from 0.92 to 2.31 percent over these levels, so the declared cross-carrier routing carries no carrier-wide A_5 covariance, and rotational covariance of a refinement limit is a premise.

These boundaries delimit one trace interpretation, one fixed-sector generator, and one conditional finite kinetic operator. The exact finite entropy, transfer, and line-graph identities hold independently of those physical attachments.

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